perceived probabilities, Thorp was able to transform the negative edge in Blackjack into a positive edge. An analogous principle would apply to a trading strategy in which it was possible to identify higher and lower probability trades.

# 16. Determining the Trade Size

What is the optimal trade size? There is a mathematically precise answer: The Kelly criterion (described in Chapter 6) will provide a higher cumulative return over the long run than any other strategy for determining trade size. The problem, however, is that the Kelly criterion assumes that the probability of winning and the ratio of the amount won to the amount lost per wager are precisely known. Although this assumption is valid for games of chance, in trading, the probability of winning is unknown and, at best, can only be estimated. If win/loss probabilities can be reasonably estimated, then the Kelly criterion can provide a starting point for determining trade size. Thorp recommends trading only half the Kelly amount (assuming win/loss probabilities can be estimated) because the penalty for overestimating the correct trade size is severe and because most people would find the volatility implied by the full Kelly amount too high for their comfort level. If win/loss probabilities can't be reasonably estimated, then the Kelly criterion can't be used.

# 17. Vary Market Exposure Based on Opportunities

Exposure levels and even the direction of exposure should vary based on opportunities and perceived relative value. For example, depending on whether stock prices appear to be cheaply or expensively priced, Claugus will vary his net exposure range from 110 percent long to 70 percent short. Varying the exposure based on opportunity can lead to significantly improved performance results.